

**Mohammed Reabai J 250701435-2025-CSE** ▾**M2****Started on** Monday, 24 August 2026, 8:06 PM**State** Finished**Completed on** Monday, 24 August 2026, 8:15 PM**Time taken** 9 mins 2 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c = 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2  int main(){
3      int n;
4      int c=0;
5      scanf("%d",&n);
6      c+=2;
7      for(int i=n/2;i<n;i++){
8          c+=2;
9          for(int j=1;j<n;j=2*j){
10             c+=2;
11             for(int k=1;k<n;k=k*2)
12                 {
13                     c+=2;
14                 }
15             }
16         }
17     printf("%d",c);
18     return 0;
19 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)